Project Title : Edge-Deployable Melanoma Detection API

Dataset : HAM10000 (Human Against Machine with 10000 training images)

Objective : Develop an optimized, low-latency computer vision API for early skin cancer detection, designed to be deployed in resource-constrained clinical environments.

Pipeline :
1) Data Ingestion

Collect images from HAM10000

Organize dataset versions using DVC

Metadata ingestion (age, sex, lesion localization)

2) Data Validation

Image integrity checks

Duplicate detection

Resolution validation

Label consistency checks

Automated validation with Great Expectations

3) Preprocessing & Feature Engineering

Resize / normalize images

Data augmentation

Color normalization

Hair artifact removal

Class imbalance handling

Metadata fusion with image features

4) Model Training

Baseline: ResNet50

Candidate models: EfficientNet, ConvNeXt

Experiment tracking: MLflow

Metrics: Accuracy, F1-score, ROC-AUC, Sensitivity / Specificity

5) Explainability

Grad-CAM heatmaps

Confidence calibration

Uncertainty estimation

6) Model Optimization

Export to ONNX

Quantization (INT8)

Pruning

Latency benchmarking

Memory footprint optimization

7) Deployment

Inference API using: FastAPI

Containerization: Docker

Target: Edge CPU / Raspberry Pi / Jetson / mobile deployment

8) Monitoring

Drift detection using Evidently AI

Inference latency monitoring

Confidence monitoring

Data quality monitoring

9) CI/CD

Automated tests

Automated retraining pipeline

Model registry / rollback

Deployment automation with GitHub Actions